

ALI NIKKHAH

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RESEARCH INTERESTS

Vision-language models; multimodal representation learning; explainable and robust AI; speech and audio intelligence; applied machine learning for healthcare and financial systems.

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major

Sept 2020 – Jul 2024

- Relevant coursework: Deep Learning, Machine Learning, Digital Signal Processing, Linear Algebra, Probability & Statistics, Algorithm Design, Digital Systems Design.

PUBLICATIONS & PREPRINTS

Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning

2025

Ali Nikkhah, [Co-authors TBD], **A. Supervisor**

arXiv preprint – in preparation

RESEARCH EXPERIENCE

University of British Columbia

Remote

Research Assistant – Vision-Language Medical Imaging

Feb 2024 – Nov 2025

- Developed vision-language architectures (**ViT**, **BLIP**, **T5**) trained with self-supervised contrastive learning to generate DICOM-compliant diagnostic reports from raw ultrasound scans; achieved SOTA **BLEU** and **ROUGE-L** on the evaluation benchmark.
- Investigated cross-modal alignment strategies to bridge the semantic gap between ultrasound image features and structured radiology report language.
- Work resulted in a preprint in preparation (see Publications).

Trinity College Dublin – EmoDub Project

Remote

ML Researcher – Emotion-Aware Voice Translation

May 2025 – Nov 2025

- Researched emotion-preserving voice translation, developing architectures that maintain both linguistic content and affective characteristics across language boundaries.
- Fused acoustic, semantic, and contextual embeddings using **wav2vec2.0** and **Whisper** to represent speaker emotion independently of lexical content, enabling emotionally faithful dubbing generation.

L3S Research Center (Leibniz University Hannover)

Remote

Research Assistant – Video Motion Classification

Apr 2023 – Feb 2024

- Designed texture-free motion classification pipelines using 3D CNNs and transformer encoders applied to dense optical flow fields, deliberately suppressing appearance cues to isolate generalizable human-dynamic representations.
- Achieved state-of-the-art classification accuracy on **UCF-101** and **HMDB-51** benchmarks; findings demonstrated that flow-based texture-agnostic features transfer more robustly across recording conditions.

Sharif University of Technology – Applied ML Lab

Tehran, Iran

Research Collaborator – Explainable AI

Jul 2025 – Sep 2025

- Investigated domain-agnostic Explainable AI (XAI) frameworks, applying advanced attention visualization techniques to probe the internal representations of large neural networks.
- Explored robustness of ML model explanations under distribution shift, contributing to lab discussions on interpretability evaluation methodology.

TEACHING & MENTORING

Sharif University of Technology

Tehran, Iran

Graduate & Undergraduate Teaching Assistant

Jan 2021 – Jan 2024

- **Foregrounded courses (graduate-level):** Deep Learning, Natural Language Processing, Data Science, Machine Learning.
- **Additional courses (undergraduate-level):** Generative Models, Engineering Probability & Statistics, Analog Electronics, Circuit Theory.
- Responsibilities across roles included designing and grading problem sets, holding weekly office hours for cohorts of 30–80 students, preparing tutorial sessions, and contributing to exam question design for graduate-level deep learning and NLP courses.
- 3 years continuous TA service across 8 distinct courses spanning both core electrical engineering and advanced AI/ML curricula.

INDUSTRY EXPERIENCE

- Digikala**
AI Engineering R&D Lead

Tehran, Iran
Nov 2024 – Aug 2025

 - Led applied research into agentic RAG architectures, implementing Knowledge-Augmented Generation (KAG), hybrid dense/sparse retrieval (RRF), and HyDE query expansion in a production system serving 5M+ daily queries.
 - Investigated multi-agent system design using LangGraph: modeled tool-calling agents as cyclic directed graphs with explicit state transitions, evaluating tradeoffs between agent autonomy and output reliability.
- Advanced Analytics Australia**
Computer Vision Engineer

Remote Contract
Sep 2025 – Apr 2026

 - Applied deep vision models (ViT, object detection, optical flow) to real-world safety compliance; investigated quantization and runtime compilation (TensorRT, TF Lite) for constrained edge deployment.
- Turquoise Digital (Firouzeh Digital)**
Senior Data Scientist & Data Engineer

Tehran, Iran
Aug 2025 – Mar 2026

 - Applied time-series transformer models to financial forecasting; designed agentic chatbot with function-calling for secure live data retrieval; built telephony audio transcription pipeline using Whisper and wav2vec2.0.
- HomaCloud**
MLOps Engineer

Tehran, Iran
May 2021 – Feb 2022

 - Orchestrated distributed ML pipelines on Kubernetes (Ray, MLflow, TensorFlow Serving); managed CI/CD and infrastructure-as-code (Terraform, ArgoCD).

TECHNICAL SKILLS

Languages: Python, C++, Go, SQL, R, Bash, TypeScript
Deep Learning & Vision: PyTorch, TensorFlow, JAX, Hugging Face, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa
Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, FAISS, ChromaDB, HyDE, KAG, Graph RAG
Data & MLOps: Spark, Airflow, Kafka, Docker, Kubernetes, Ray, Triton, vLLM, MLflow, W&B
Languages (Human): Persian (Native), English (C2 – IELTS 8.0), Arabic (Professional), French (Beginner)